

David Matzek

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CONTACT

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*Most work is in BMW's private GitHub Enterprise;
walkthrough on request.*

🌐 davidmatzek.github.io

LANGUAGES

German

Native



English

C1 – Oxford, UK & Chicago, USA



Spanish

B1 – Monterrey, Mexico



KEY COMPETENCIES

- Taking ownership & accountability
- Proactive, solution-driven mindset
- Involving partners and teams early
- Leading without formal authority
- Building early, failing fast
- Executive-level communication
- Decisive under budget pressure
- Mentoring and enabling others

SENIOR DATA & SYSTEMS ARCHITECT

AI automation for data-model governance — with BMW since 2010, 9 years in engineering

Profile

Senior data architect, with BMW since 2010: 9 years in engineering, 5 of them in data architecture. I own BMW's vehicle-data model and automated its governance with AI agents. I take on critical projects when budgets and teams shrink. Next step: building AI-driven data systems in growing markets. Seeking senior or lead roles in Munich (hybrid, permanent); I live in Nördlingen.

PROFESSIONAL HIGHLIGHTS.

- ✓ **~€350k per year cost avoided.** An AI-agent system replaced two consultants and one internal position (~2.5 FTE); daily effort for a 240-consumer data model dropped from ~24 h to 35 min (estimated).
- ✓ **20 workdays.** Time to deliver a self-service feature that had been open for 6 years.
- ✓ **Owner of BMW's VSS data model** for 180+ applications; steering BMW's liaison to COVESA and feeding implementation experience back.
- ✓ **3 patent filings as inventor**, one granted in the US, from in-car function development.

PROFESSIONAL POSITIONS.

01/2025 –
today

Data Steward & Data Model Owner | BMW Group

Vehicle Data Assets

- Own BMW's VSS vehicle-data model (Vehicle Shadow), consumed by 180+ backend applications; steward of petabyte-scale historized data.
- Replaced two consultants and one internal position with 7 AI agents + a self-service "VSS Explorer" UI — ~€350k/year avoided — automating PR triage, CI-fixing, EU Data Act classification and releases.
- Sustained weekly releases after the 01/2026 budget cut (estimated ~24 h → 35 min/day); driving the .vspec → GraphQL (S2DM) model migration.

01/2023 –
12/2024

Lead Data Designer | BMW Group

Vehicle Data Infrastructure

- Established an early-phase data-design discipline across a 60+ ECU board net.
- Set E/E (physical bus) requirements so needed data reaches in-car collectors by design.

11/2021 –
12/2022

E/E Data Architect | BMW Group

Data Strategy, Connected Vehicle

- Middleware-based (ADAPT) standardization reducing model fragmentation across platforms.
- Variant management and test automation for cross-derivative data.

04/2017 –
11/2021

Specialist Function Development | BMW Group

Intelligent Personal Assistant

- Function owner, customer-facing in-car function idea → go-live: a ~€3M program over 3 years across the full V-model lifecycle.
- 6 in-car functions to series — two in 2019, four in 2021; inventor on patent filings (page 2).

SOFTWARE & TOOLS

AI & automation

GitHub Copilot, MCP, agentic workflows

Engineering

Python, GraphQL, YAML, Git & GitHub, VS Code, CI/CD, Cloud

Process & quality

Jira, Confluence, ALM Octane

Collaboration

MS Office (Word, Excel, PowerPoint)

QUALIFICATIONS

- Internal first aider
- Internal driving licenses

INTERESTS

- Digitalization & entrepreneurship
- Endurance sports
- Personal development
- Education & mentoring
- Cooking & coffee

SELECTED CASE — AI-AGENT AUTOMATION

01/2026 –
today

VSS Data-Model Governance, Automated

BMW Group | Vehicle Shadow

- 7 purpose-built agents (~15 skills) orchestrate PR triage, CI-fixing, auto-rebase, EU Data Act classification, changelog and version bumps, and release-board sorting.
- “VSS Explorer” self-service UI lets non-technical contributors request signals without git/PR skills; a backend pipeline auto-generates compliant PRs.
- Cost-aware LLM tooling: description rewriter benchmarked across 3 model tiers for best cost/quality.
- COVESA: BMW’s liaison department (set up for compliance reasons) works under my guidance; I give implementation feedback to the VSS community.
- Impact: ~€350k/year (~2.5 FTE: two consultants and one internal position) of governance cost avoided and consultant dependency ended, for 45+ users at a weekly release cadence.

PATENT FILINGS (INVENTOR).

WO2020083962A1

Apparatus and control unit for automating a state change of a window pane of a vehicle.

Sole inventor • published 04/2020 • filed in DE, CN, US (US granted)

WO2020011655A1

Method and system for detecting data in vehicles.

Sole inventor • published 01/2020 • filed in DE, CN, US

DE102018209967A1

Method for automatically configuring a function in a vehicle.

Co-inventor • published 12/2019 • filed in DE

EDUCATION

10/2018 –
10/2021

M.Sc. Systems Engineering

University of Applied Sciences Munich

- In parallel to professional position at BMW.
- Master thesis: influence of frontloading on software-development efficiency.

10/2013 –
01/2017

B.Eng. Mechatronics

University of Applied Sciences Esslingen

- Focus on automotive electronics; BMW SpeedUp scholarship (future-talent program).
- Bachelor thesis (07/2016 – 01/2017): external perception of autonomous systems.

08/2015 –
01/2016

Exchange Semester

Tecnológico de Monterrey, Mexico

- Cross-cultural experience.

09/2010 –
07/2013

Apprenticeship Mechatronics (IHK)

BMW Group, Munich

- In-house project lead with multi-million-euro scope.